Lab - Introduction to Amazon DynamoDB

Lab Overview

Amazon DynamoDB is a fast and flexible NoSQL database service for all applications that need consistent, single-digit millisecond latency at any scale. It is a **fully managed database** and supports both document and key-value data models. Its flexible data model and reliable performance make it a great fit for mobile, web, gaming, ad-tech, IoT, and many other applications.

In this lab, you create a table in Amazon DynamoDB to store information about a music library. You then query the music library and, finally delete the DynamoDB table.

Objectives

After completing this lab, you will be able to:

- Create an Amazon DynamoDB table.
- Enter data into an Amazon DynamoDB table.
- Query an Amazon DynamoDB table.
- Delete an Amazon DynamoDB table.

Start lab

1. To launch the lab, at the top of the page, choose **Start Lab**.

Caution: You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose **Open Console** .

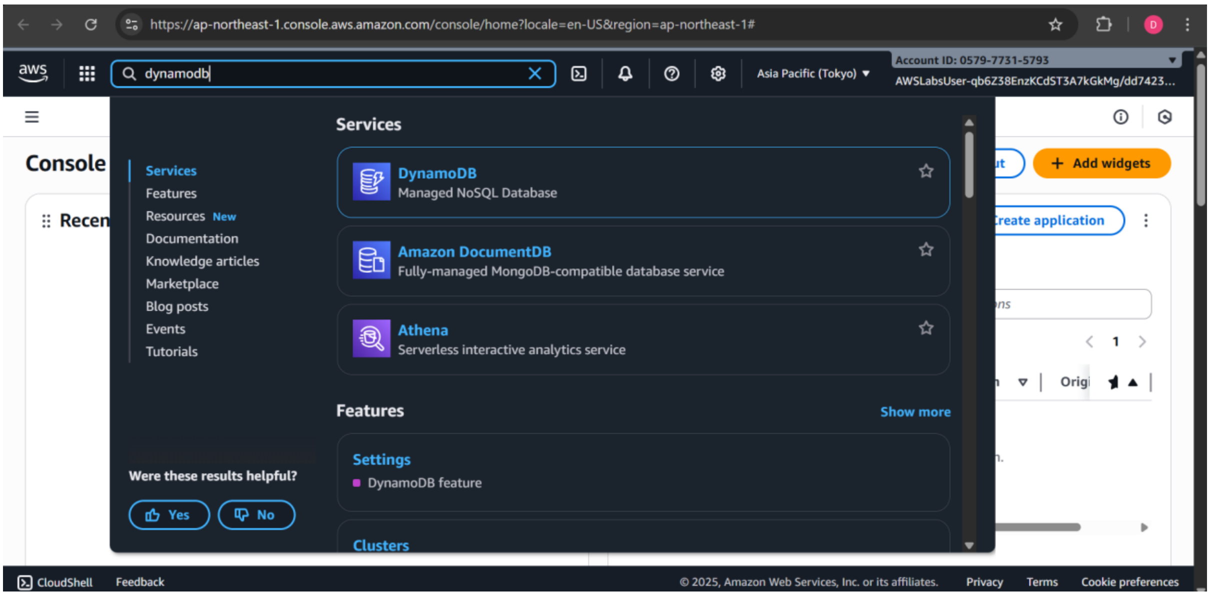
You are automatically signed in to the AWS Management Console in a new web browser tab.

Warning: Do not change the **Region** unless instructed.

Task 1: Create a New Table

In this task, you create a new table in DynamoDB named *Music*. Each table requires a Primary Key that is used to partition data across DynamoDB servers. A table can also have a Sort Key. The combination of Primary Key and Sort Key uniquely identifies each item in a DynamoDB table.

3. At the top of the AWS Management Console, in the search bar, search for and choose DynamoDB



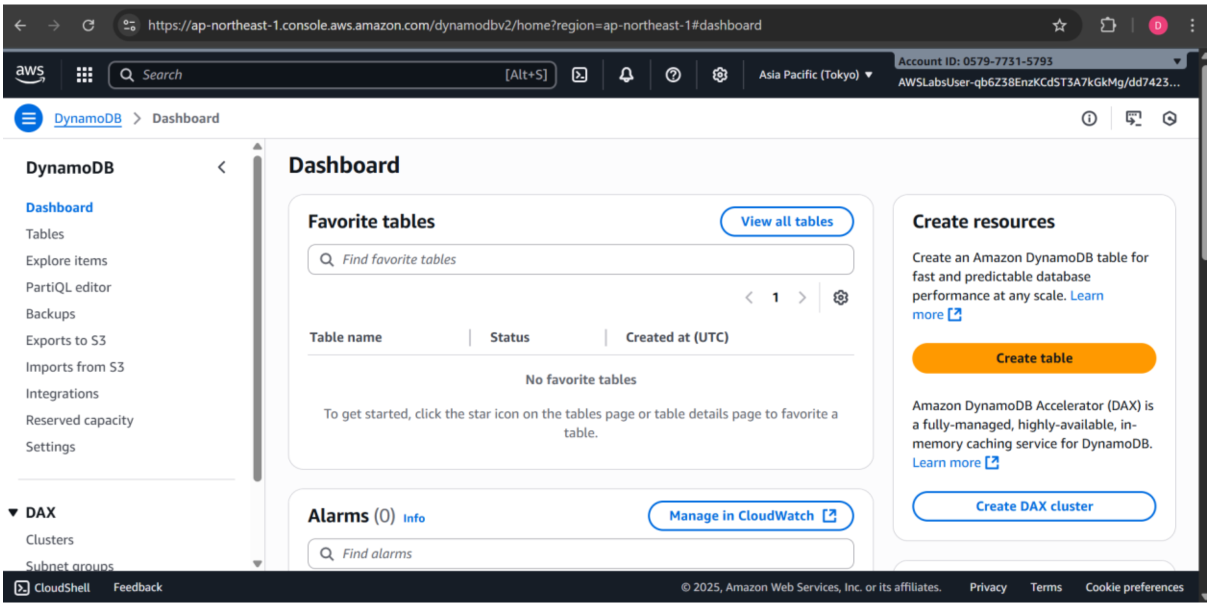
4. On the **Amazon DynamoDB** getting started page, choose **Create table**.

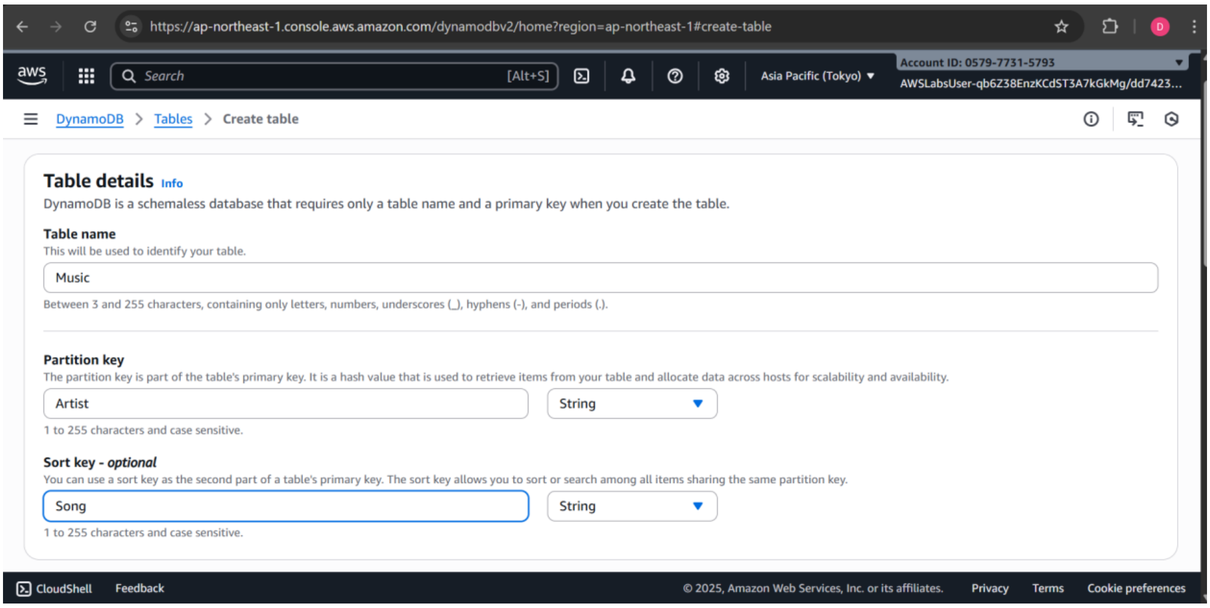
On the **Create table** page, configure the following options:

In the **Table details** section:

- For **Table name**, enter Music
- For **Partition key**, enter Artist and choose **String** from the dropdown menu.
- For **Sort key - *optional***, enter Song and choose **String** from the dropdown menu.

Your table uses default settings for indexes and provisioned capacity.





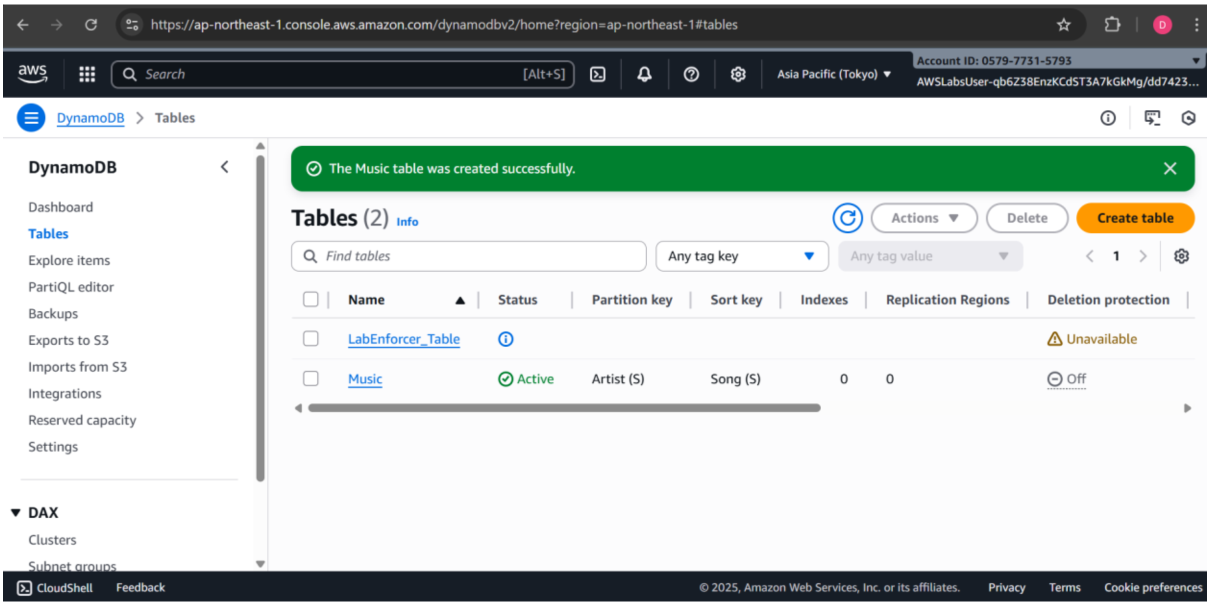
5. Choose **Create table**.

Expected output: Initial message.

A Creating the Music table. It will be available for use shortly. message is displayed on top of the screen.

6. Wait for your table to be created and its status to be Active.

Congratulations! You have successfully created a new table named *Music* in DynamoDB.



ask 2: Add Data

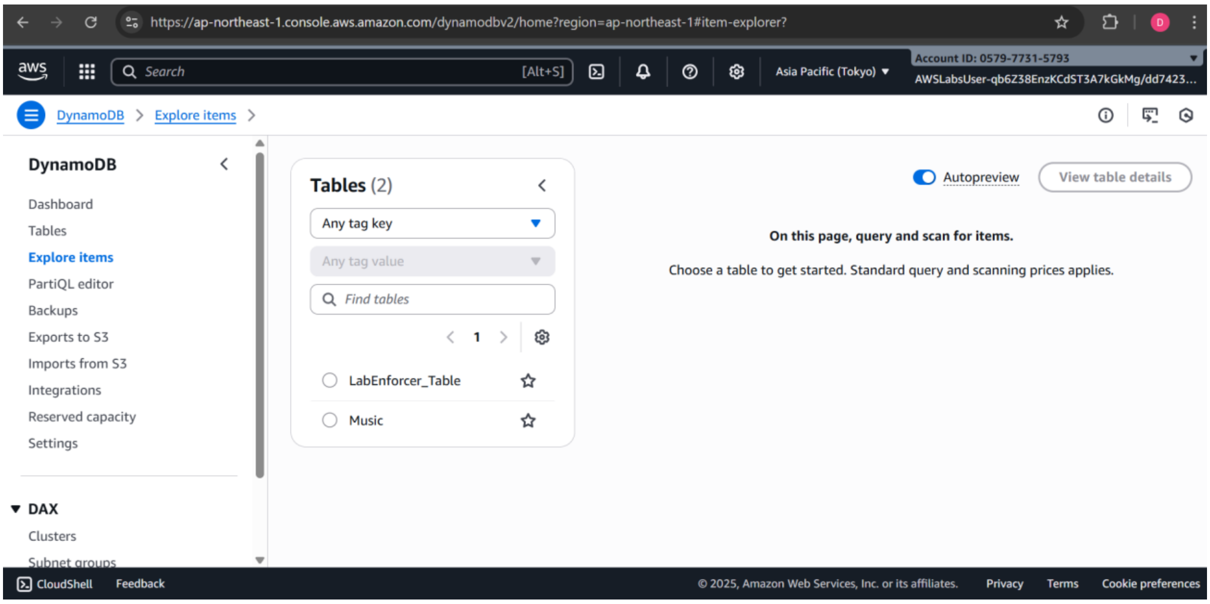
In this task, you add data to the *Music* table. A **table** is a collection of data on a particular topic.

Each table contains multiple **items**. An item is a group of attributes that is uniquely identifiable among all of the other items. Items in DynamoDB are similar in many ways to rows in other database systems. In DynamoDB, there is no limit to the number of items you can store in a table.

Each item is composed of one or more **attributes**. An attribute is a fundamental data element, something that does not need to be broken down any further. For example, an item in a *Music* table contains attributes such as Song and Artist. Attributes in DynamoDB are similar columns in other database systems, but each item (row) can have different attributes (columns).

When you write an item to a DynamoDB table, only the Primary Key and Sort Key (if used) are required. Other than these fields, the table does not require a schema. This means that you can add attributes to one item that may be different to the attributes on other items.

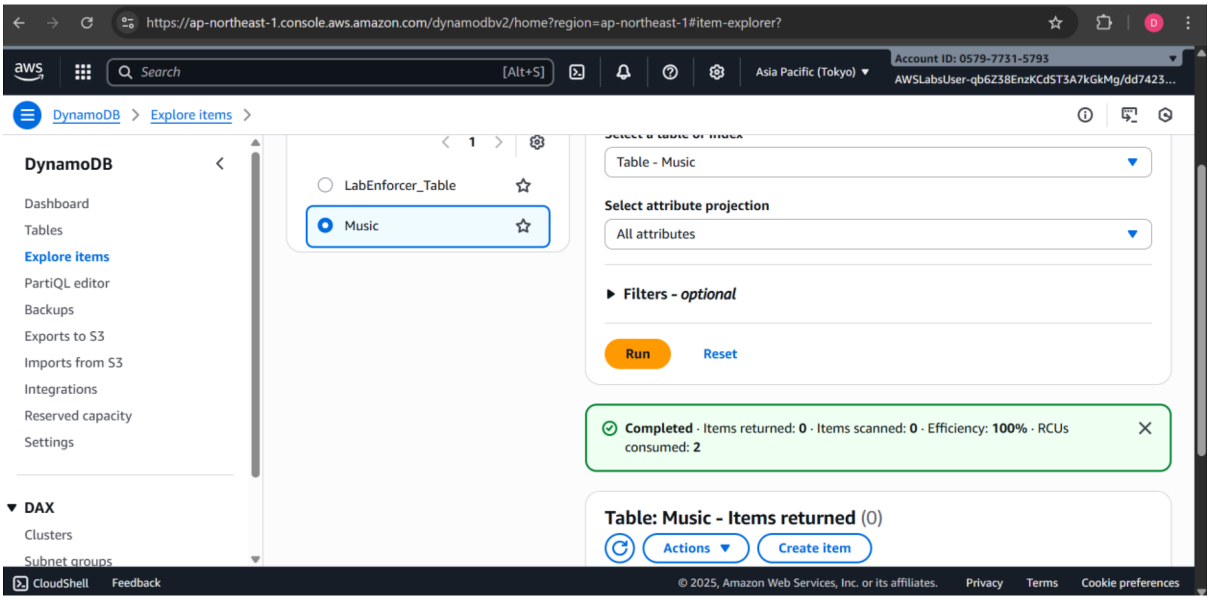
7. In the left navigation pane, choose **Explore items**.



8. Select **Music** table.

9. Choose **Create item**.

On the **Create item** page, configure the following options:



10. For **Artist** Value, enter Pink Floyd
11. For **Song** Value, enter Money

These are the only required attributes, but you also add additional attributes.

12. To create an additional attribute, choose **Add new attribute** .
13. In the drop-down list, select **String**.

A new *string* attribute row gets added.

14. For the new attribute, enter:

- For **Attribute name**, enter Album
- For **Value**, enter The Dark Side of the Moon

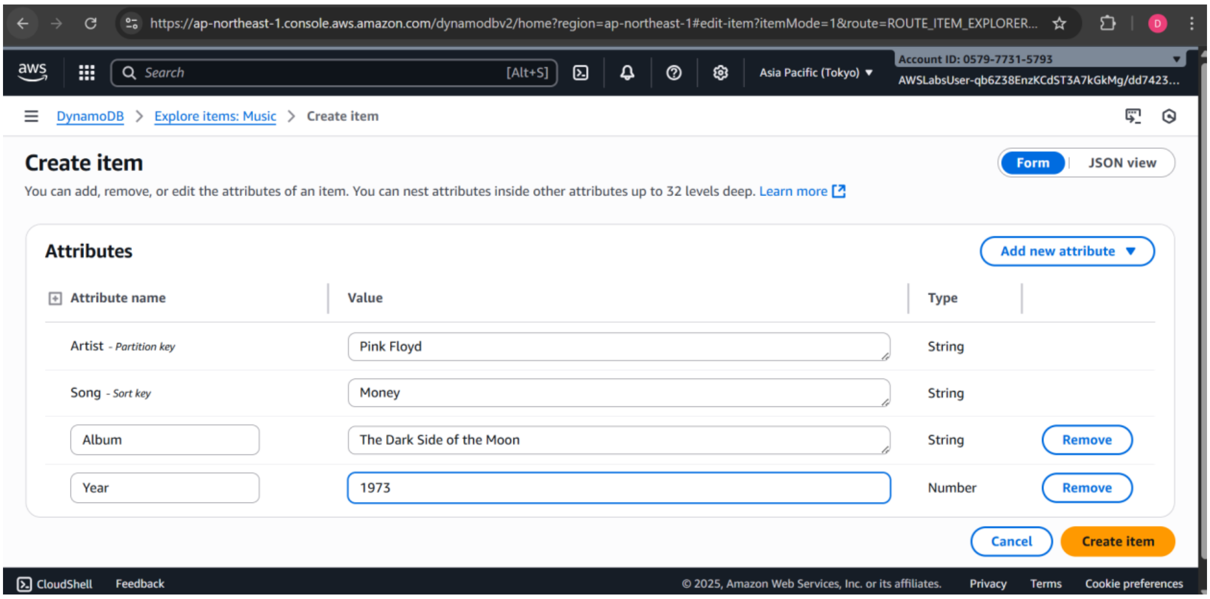
15. Create another new attribute by choosing **Add new attribute** .
16. In the drop-down list select **Number**.

A new *number* attribute gets added.

17. For the new attribute, enter:

- For **Attribute name**, enter Year
- For **Value**, enter 1973

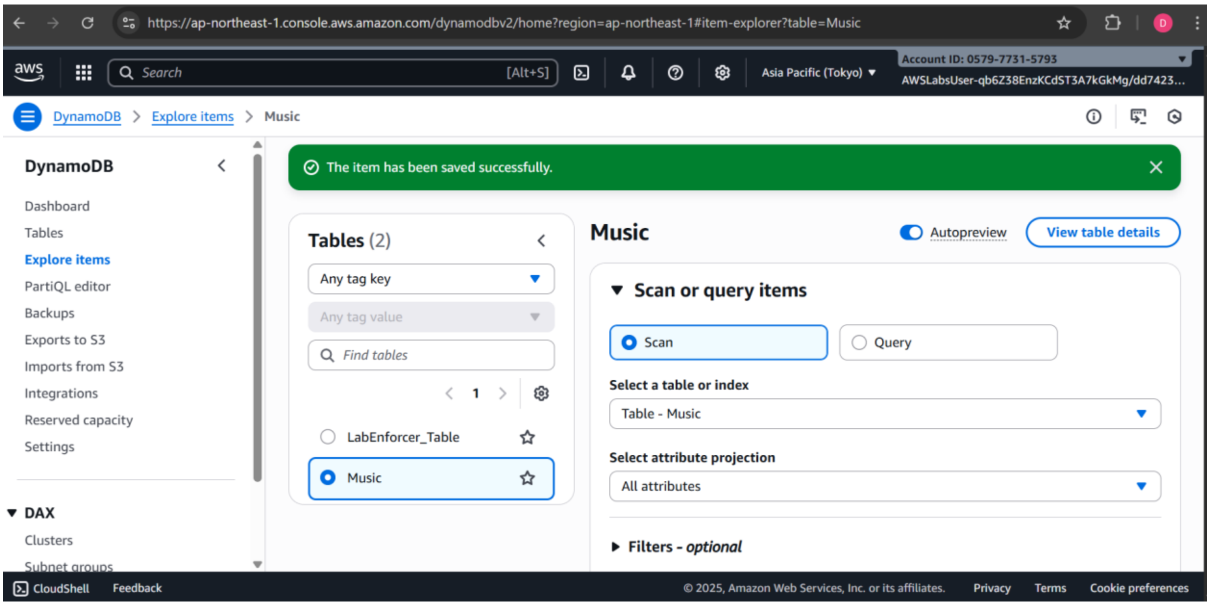
18. Choose **Create item** to store the new item with its four attributes.



Expected output:

A The item has been saved successfully. message is displayed on top of the screen.

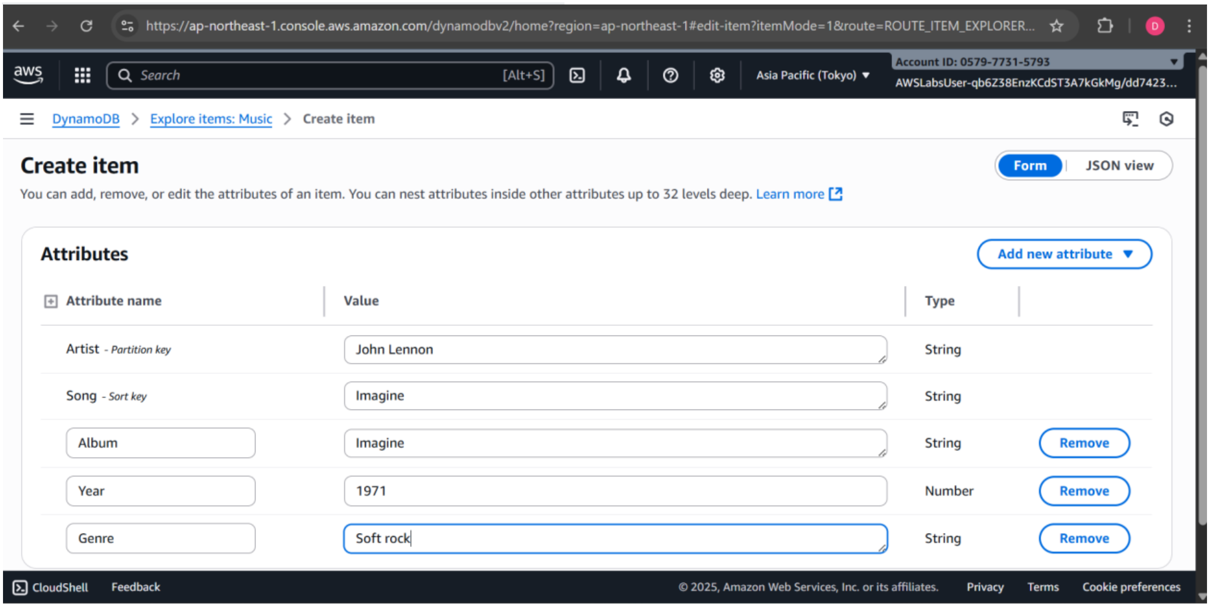
The item now appears in the console.



19. Create a second item, using these attributes:

Attribute Name	Attribute Type	Attribute Value
Artist	String	John Lennon
Song	String	Imagine
Album	String	Imagine
Year	Number	1971

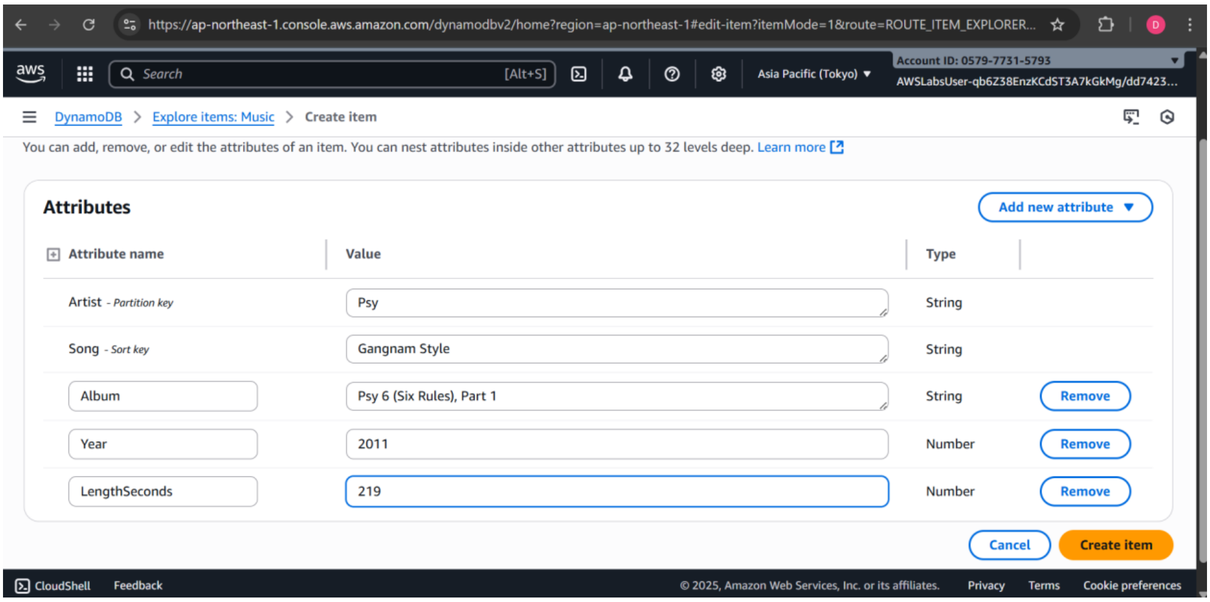
Attribute Name	Attribute Type	Attribute Value
Genre	String	Soft rock



Note that this item has an additional attribute called *genre*. This is an example of each item being capable of having different attributes without having to pre-define a table schema.

20. Create a third item, using these attributes:

Attribute Name	Attribute Type	Attribute Value
Artist	String	Psy
Song	String	Gangnam Style
Album	String	Psy 6 (Six Rules), Part 1
Year	Number	2011
LengthSeconds	Number	219



Once again, this item has a new *LengthSeconds* attribute identifying the length of the song. This demonstrates the flexibility of a NoSQL database.

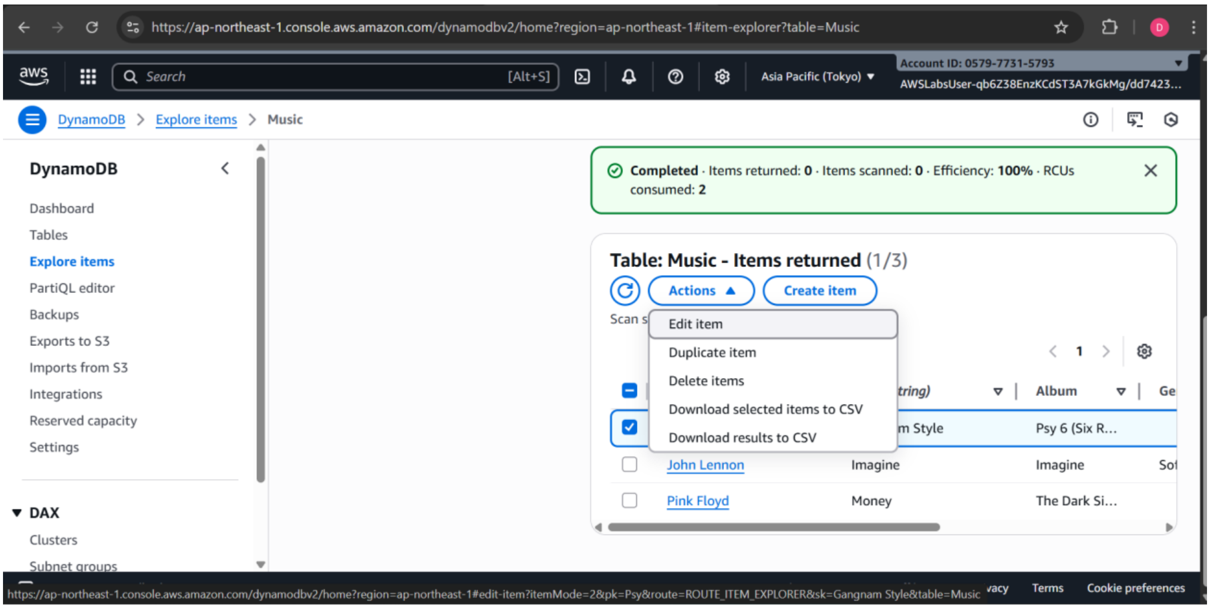
There are also faster ways to load data into DynamoDB, such as using AWS Data Pipeline, programmatically loading data or using one of the free tools available on the Internet.

Congratulations! You have successfully added data to the *Music* table.

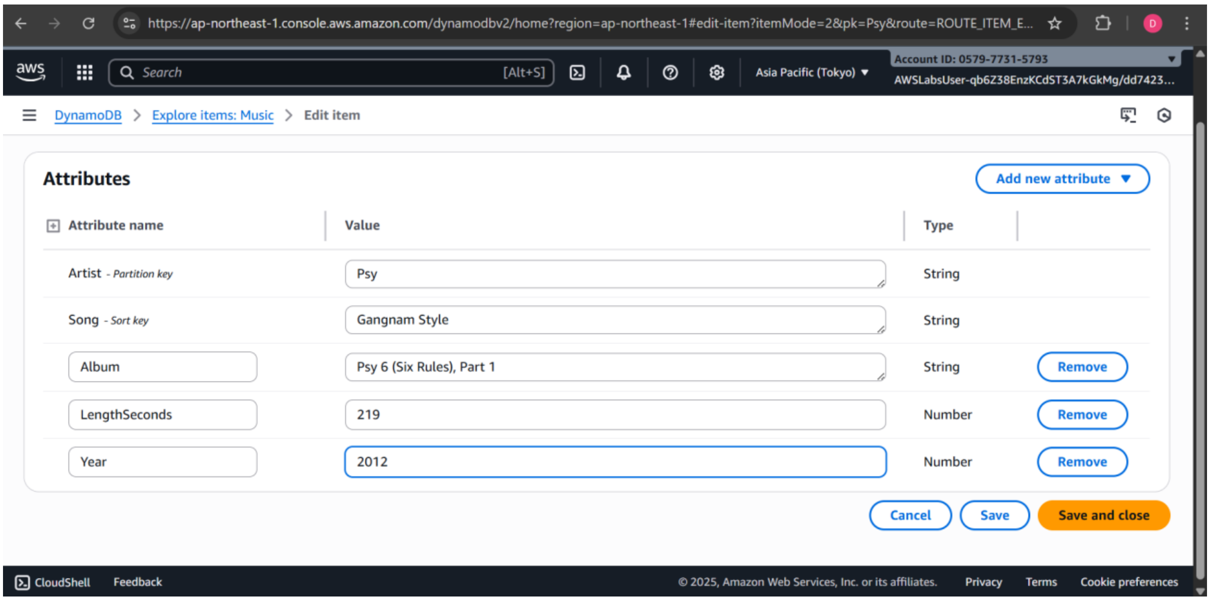
Task 3: Modify an Existing Item

In this task, you modify an existing item.

21. From the list of items, select **Psy**.
22. Choose **Actions** .
23. Choose **Edit item**.



24. Change the **Year** from *2011* to *2012*.



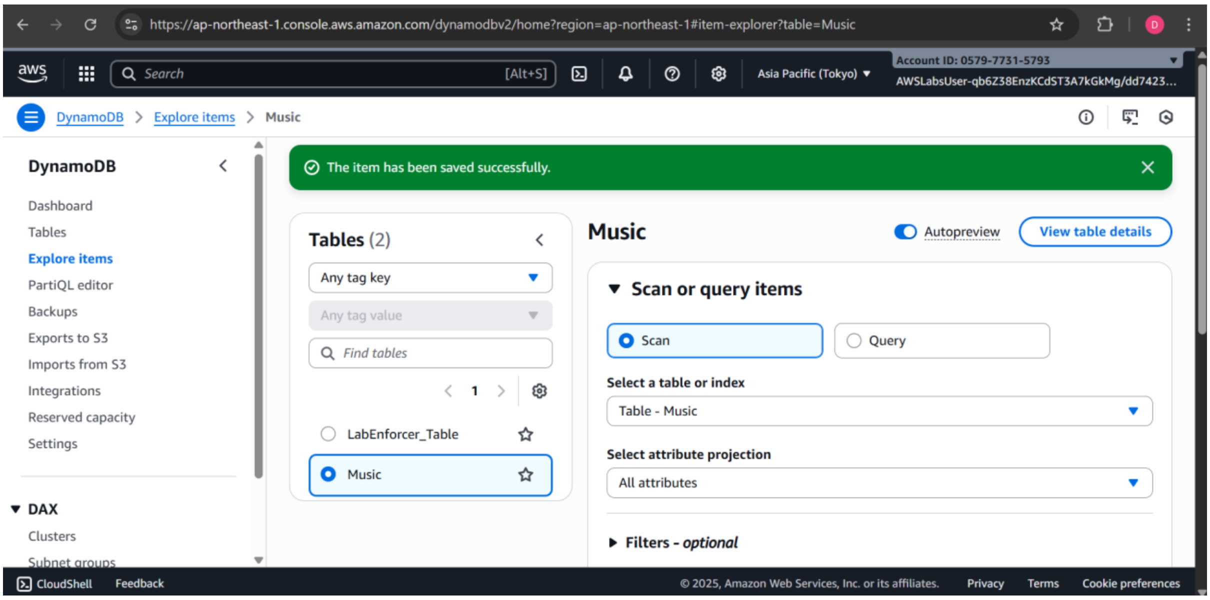
25. Choose **Save changes**.

Expected output:

A The item has been saved successfully. message is displayed on top of the screen.

The item is now updated.

Congratulations! You have successfully updated an item within the *Music* table.

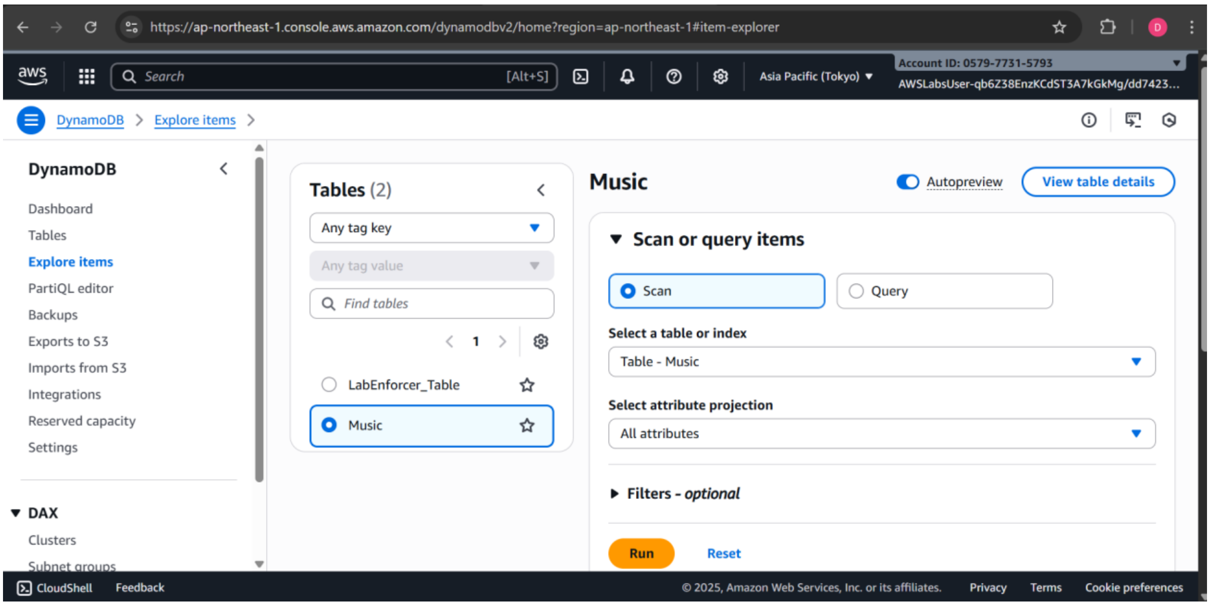


Task 4: Query the Table

In this task, you query the *Music* table. There are two ways to query a DynamoDB table: *Query* and *Scan*.

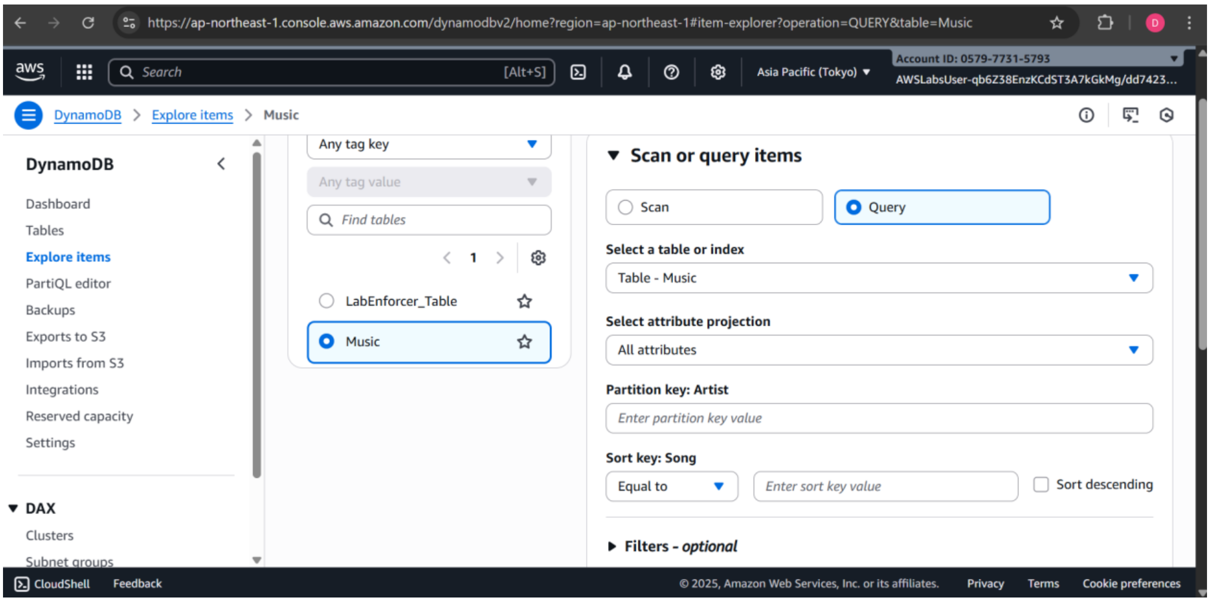
A **query** operation finds items based on Primary Key and optionally Sort Key. It is fully indexed, so it runs very fast.

26. In the left navigation pane, choose **Explore items**.
27. Select **Music** table.
28. Expand **Scan or query items** to query or scan items.



29. Select **Query**.

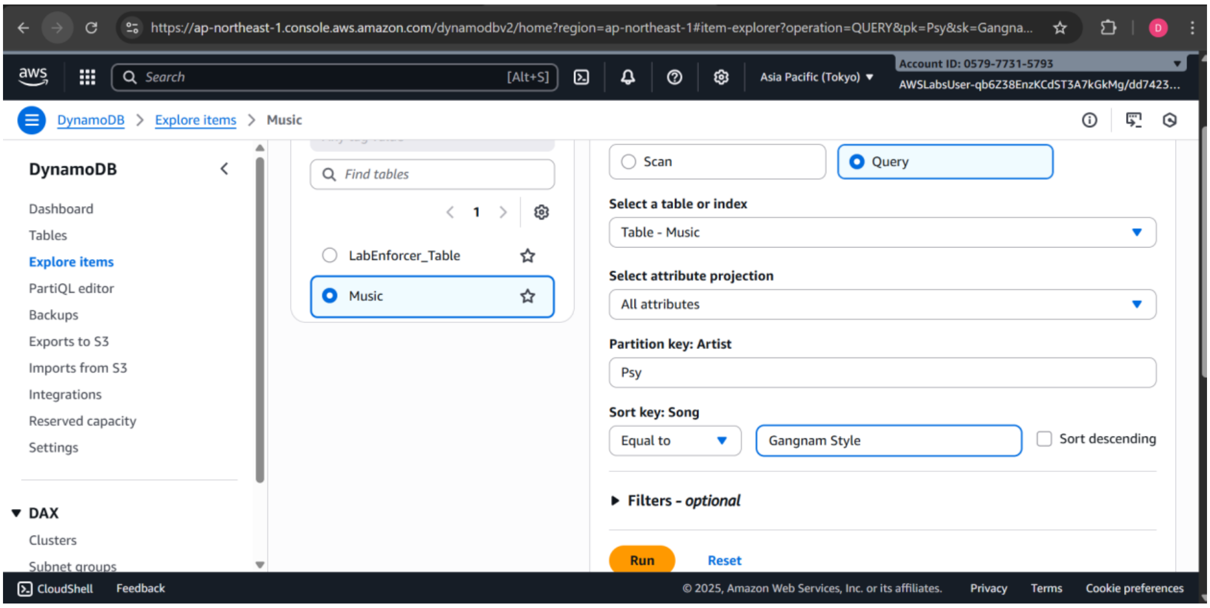
Fields for the Partition Key (which is the same as Primary Key) and Sort Key are now displayed.



30. Configure the following for Partition and Sort keys

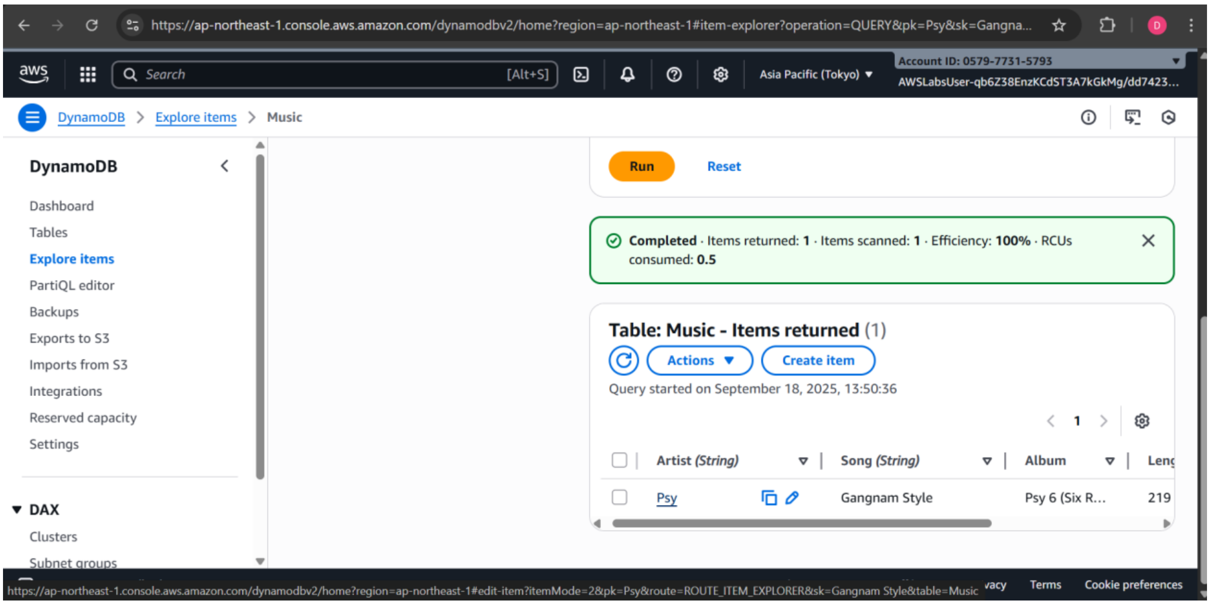
- For **Artist (Partition key)**, enter Psy
- For **Song (Sort key)**, select **Equal to** from the dropdown menu and enter Gangnam Style

31. Choose **Run**.



Expected output:

The song quickly appears in the list. A *query* is the most efficient way to retrieve data from a DynamoDB table.



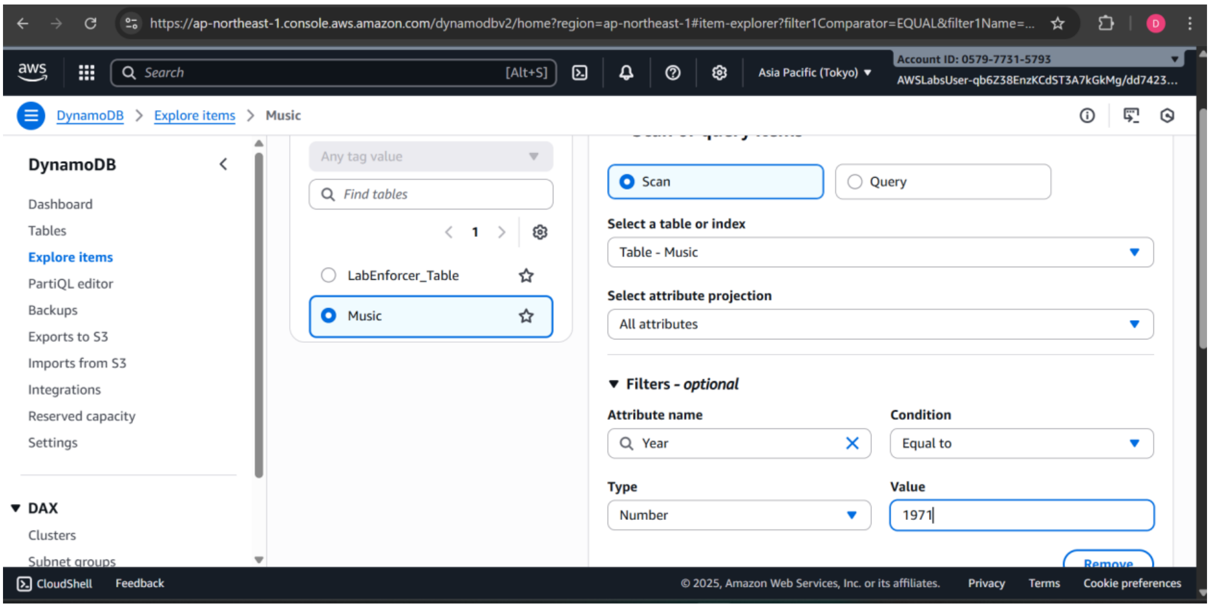
Alternatively, you can *scan* for an item. This involves looking through *every item in a table*, so it is less efficient and can take significant time for larger tables.

32. Select **Scan**.

33. Expand the **Filters** section, and configure the following:

- For **Attribute name**, enter Year

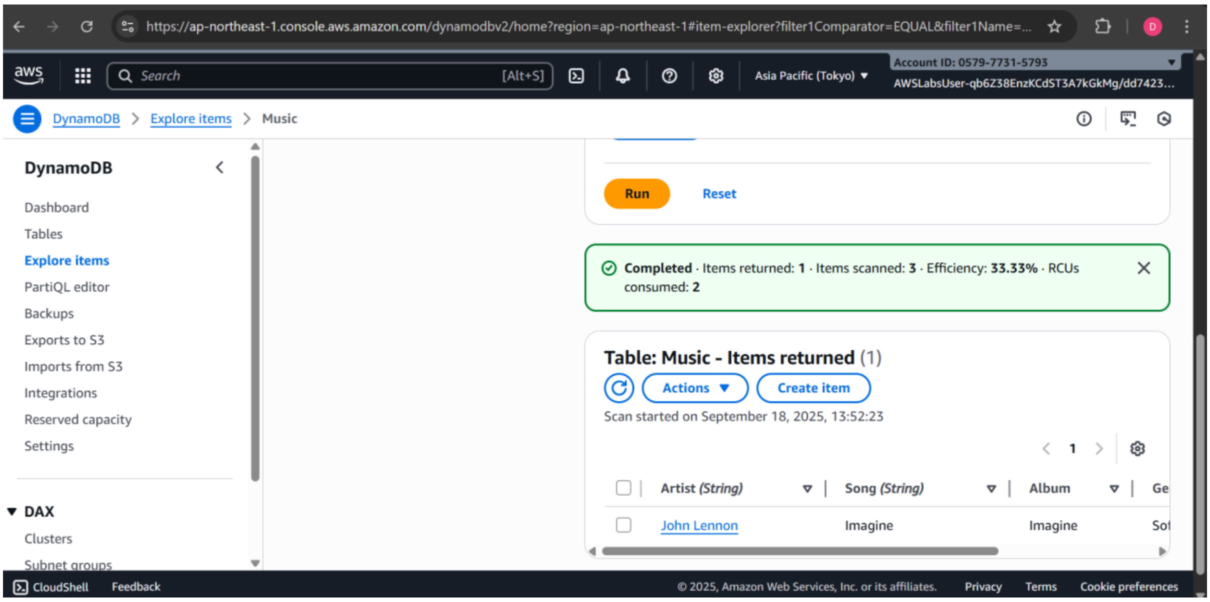
- For **Type**, select **Number** from the dropdown menu.
- For **Condition**, select **Equal to** from the dropdown menu.
- For **Value**, enter 1971



34. Choose **Run**.

Expected output:

Only the song released in 1971 is displayed.

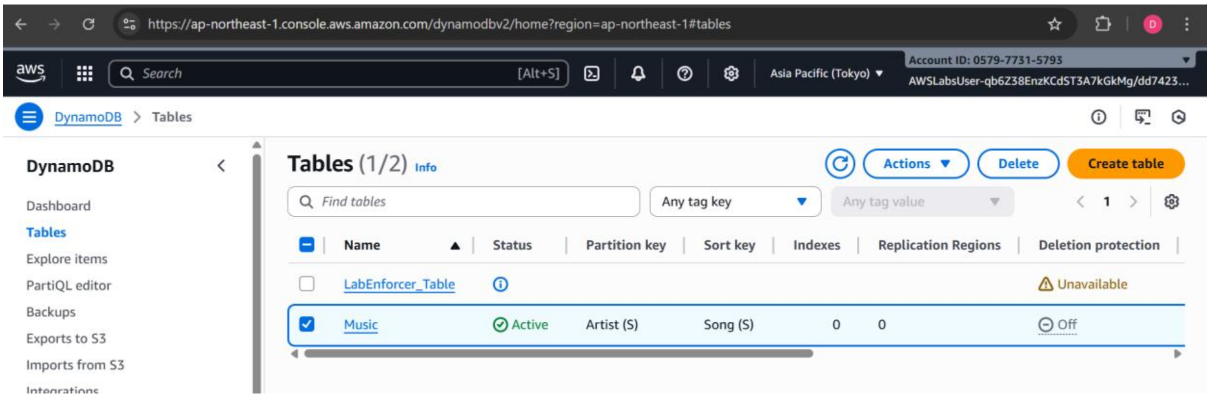


Congratulations! You have successfully used the *query* and *scan* operations to query the *Music* table.

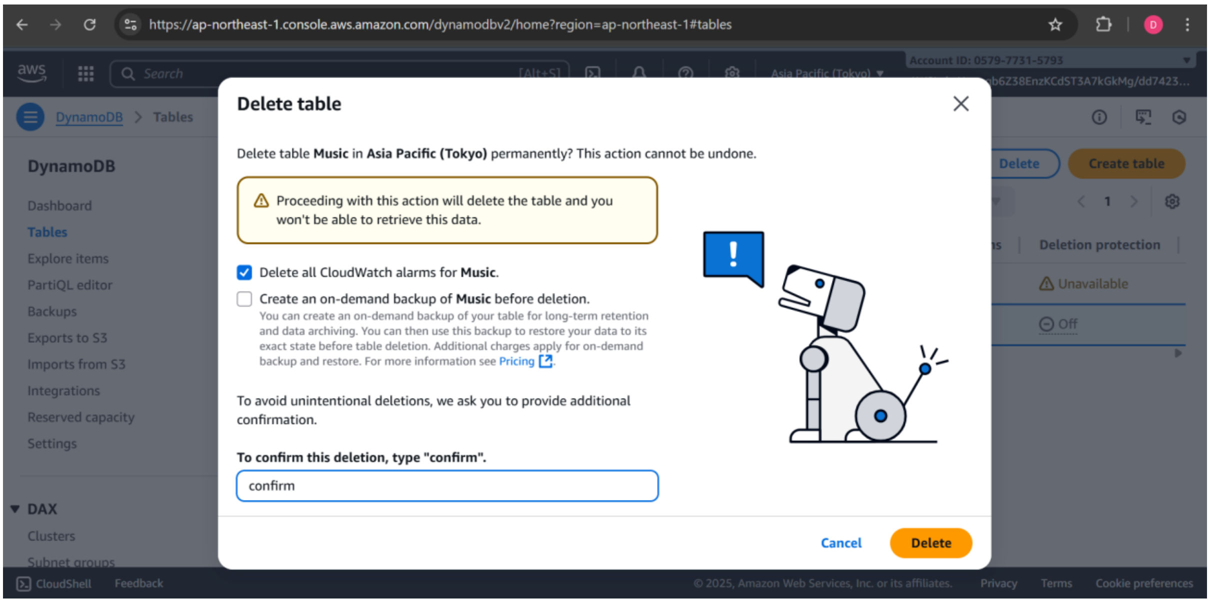
Task 5: Delete the Table

In this task, you delete the *Music* table, which also deletes all the data in the table.

35. In the left navigation pane, choose **Tables**.
36. Select **Music** table.



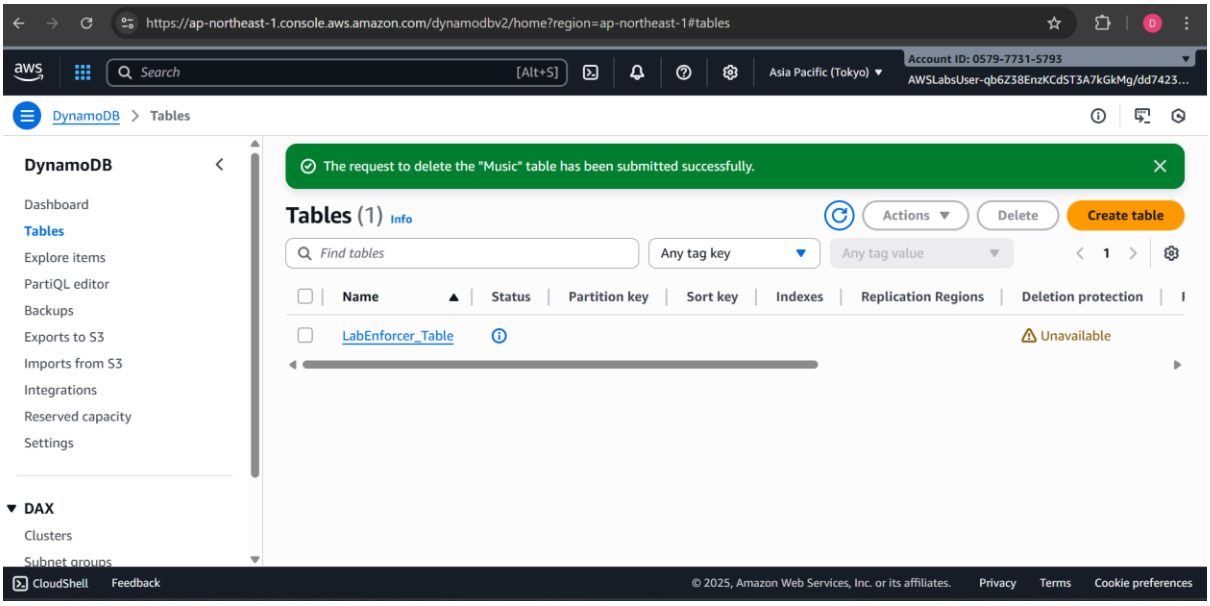
37. Choose **Delete**.
38. On the **Delete table** pop-up window, locate the **Enter "confirm" to agree.** textbox and enter confirm
39. Choose **Delete**.



Expected output: Initial message.

A The request to delete the "Music" table has been submitted successfully. message is displayed on top of the screen.

40. Use the refresh option to confirm the table deletion.



Congratulations! You have successfully deleted the *Music* table.

Conclusion

Congratulations! You have now successfully learned how to:

- Create an Amazon DynamoDB table.
- Enter data into an Amazon DynamoDB table.
- Query an Amazon DynamoDB table.
- Delete an Amazon DynamoDB table.

End lab

Follow these steps to close the console and end your lab.

41. Return to the **AWS Management Console**.
42. At the upper-right corner of the page, choose **AWSLabsUser**, and then choose **Sign out**.
43. Choose **End Lab** and then confirm that you want to end your lab.

Additional Resources

- [Amazon DynamoDB Documentation]